Big Data Analytics
CISELEC 501A-CS41S1

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Laboratory Activity #04: CRISP-DM | Data Preparation with WEKA (Feature Selection, Instance Selection, and Feature Extraction)
Total Points: 45

OBJECTIVE: This laboratory exercise is designed to provide an in-depth exploration of the Data Preparation phase of CRISP-DM framework, focusing on three essential tasks: Feature Selection, Instance Selection, and Feature Extraction. These activities aim to improve data quality, reduce noise and redundancy, and enhance the overall eUectiveness of subsequent modeling and analysis activities, ensuring that the resulting insights are both accurate and meaningful within the context of the defined problem.

By the end of this activity, students will be able to:

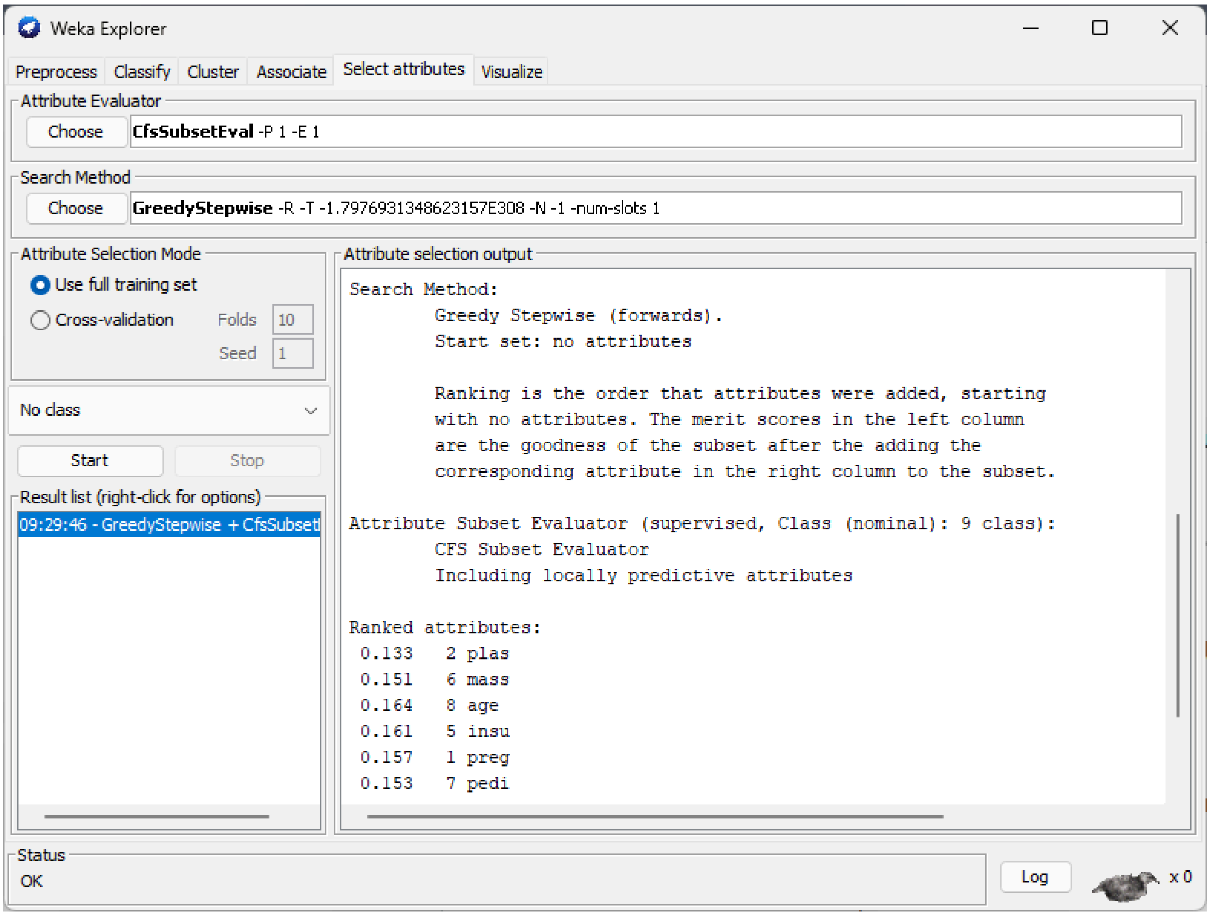
- Understand the role of data preparation in the CRISP-DM lifecycle.
- Apply WEKA tools to select relevant features that improve model performance.
- Perform instance selection to handle noisy or irrelevant data points.
- Transform features to improve consistency and readiness for machine learning algorithms.

Task 1: Feature Selection

Instructions:

Feature selection helps reduce dimensionality by retaining only the most relevant attributes.

1. Launch WEKA Explorer.
2. Load the provided dataset – diabetes.arff.
    a. Press the “Open file...” button
    b. Locate the file
    c. Press the “Open” button
3. Once the file is loaded, navigate to the Select attributes tab.
4. Choose an Attribute Evaluator (CfsSubsetEval) and a Search Method (e.g., GreedyStepwise).
5. Double-click the GreedyStepwise input box to open the weka.gui.GenericObjectEditor pane, ensure that the generateRanking attribute is set to “TRUE”, and then press “OK” to complete the configuration.
6. For the Attribute Selection Mode section, retain the default value Use full training set.
7. Run the feature selection process by pressing the “Start” button, then review the retained attributes in the Attribute Selection Output pane under the Ranked attributes: section..



Reflection:

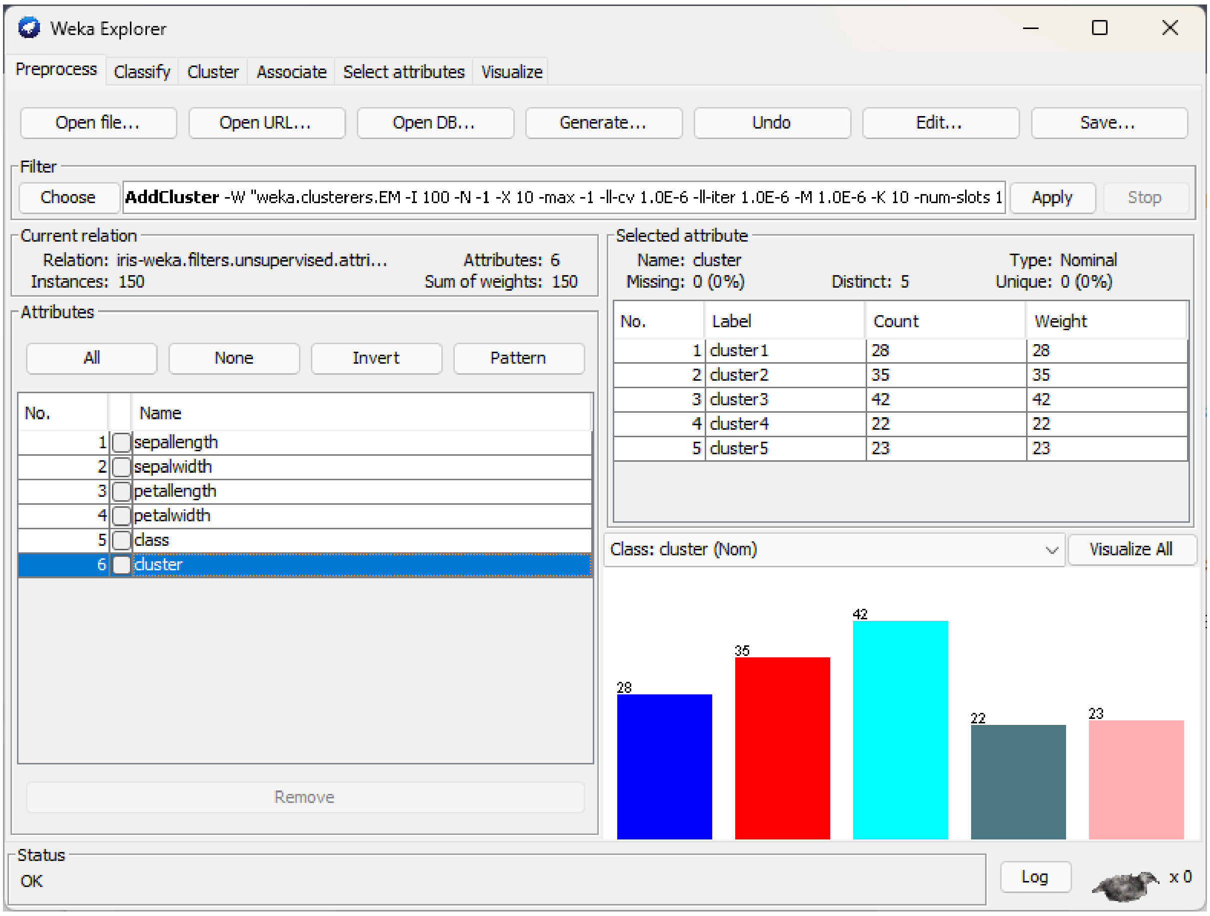
- Which features were selected? (**5pts**)
  - There are 8 features that were selected:
    - plas
    - mass
    - age
    - insu
    - preg
    - pedi
    - skin
    - pres
- What are the top 3 attributes detected by the CfsSubsetEval algorithm based on their Ranked values? (**5pts**)
  - Top 3 attributes based on their rank value is:
    - age - 0.164
    - insu - 0.161
    - preg - 0.157
  - This indicates that on our feature selection, **age**, **insu** and **preg** are the top 3 influential attributes that hold a strong correlation on the attribute or targeted “class” named class. This helps us think that the top 3 ranked values have such a strong influence whether an object is tested positive or negative.

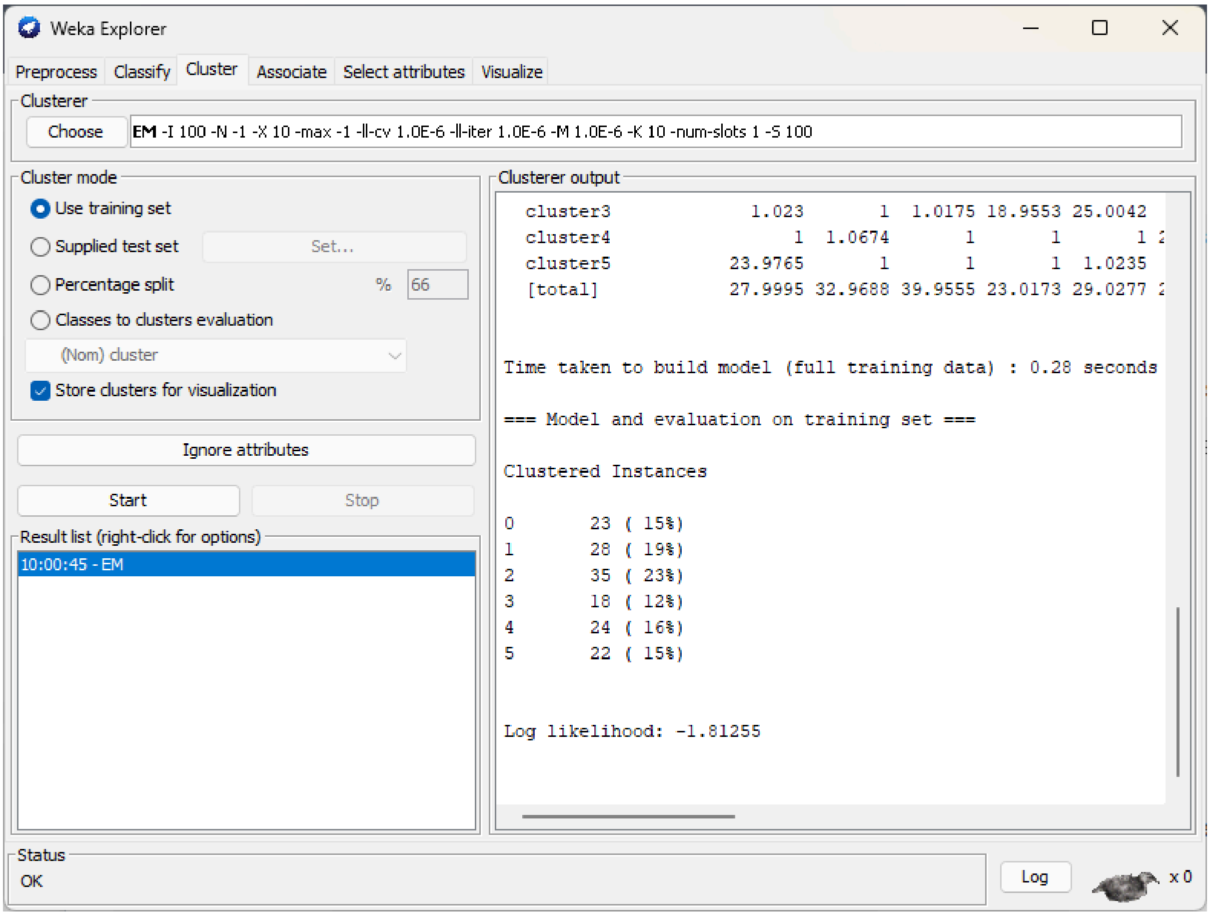
Task 2: Instance Selection

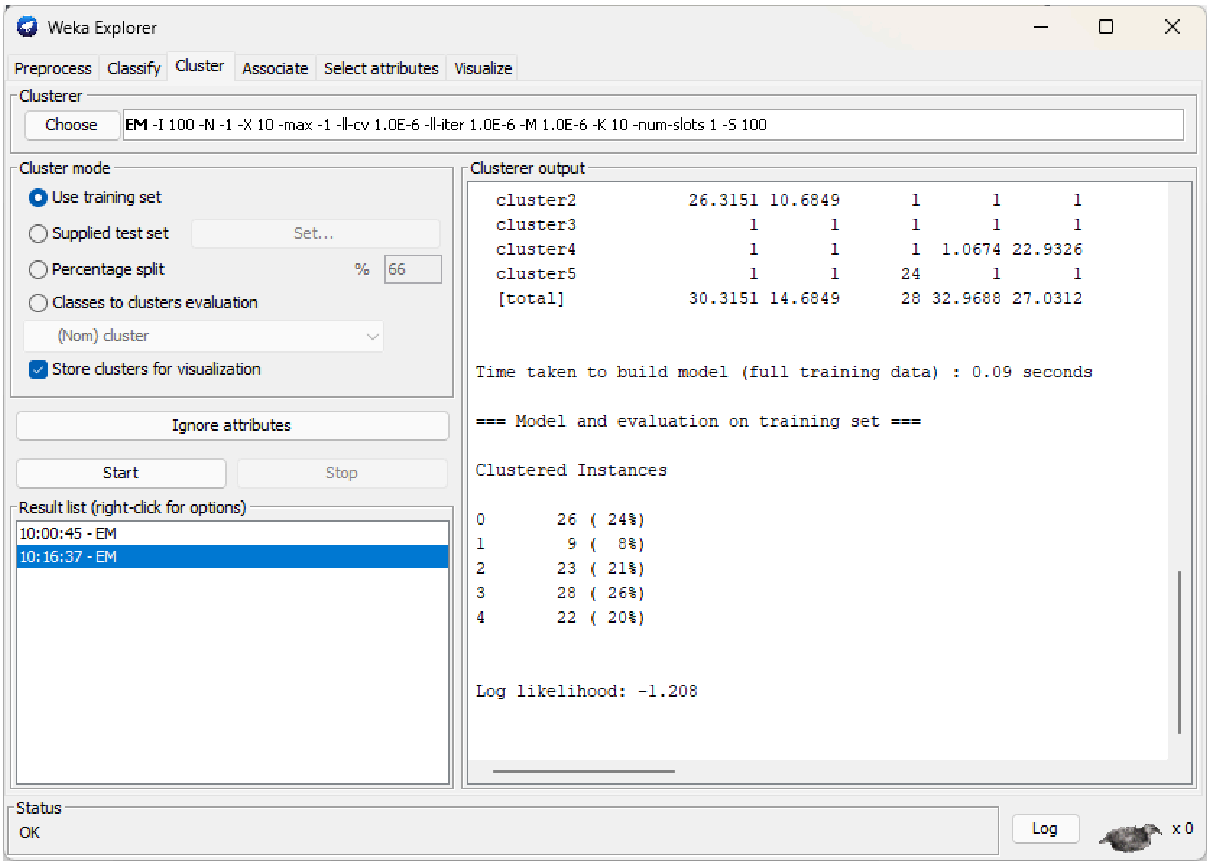
Instructions:

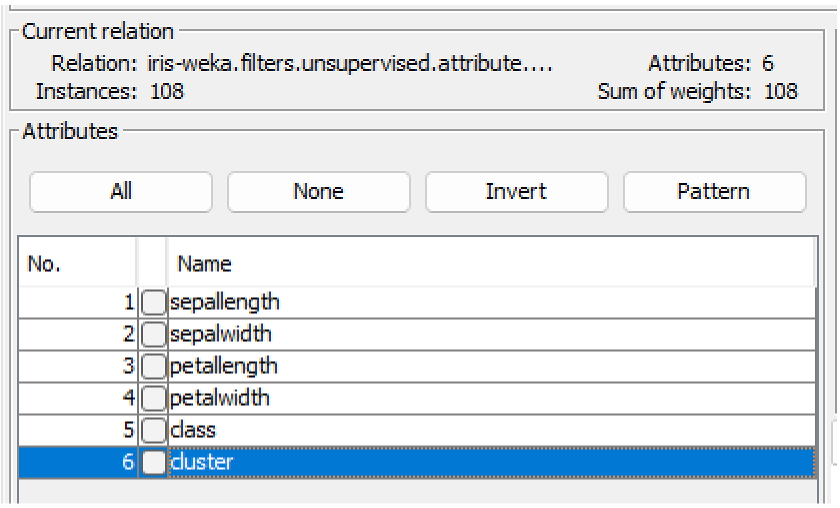
Instance selection involves filtering or removing noisy, irrelevant, or duplicate data points to improve model quality.

1. Navigate to the Preprocess tab in WEKA's Explorer interface and load the dataset iris.arff
2. Click the Choose button under the Filter section and select:
    a. unsupervised → attribute → AddCluster (this will add a cluster assignment attribute to each instance).
3. In the Clusterer option for AddCluster, choose:
    a. weka.clusterers.EM
4. Configure EM parameters by clicking the clusterer name:
    a. Leave Number of clusters set to -1 (automatic) to let EM determine the optimal number of clusters.
    b. Leave other parameters as default for an initial run.
    c. Click OK to save the settings.
5. Click Apply to run the filter. A new attribute (e.g., cluster) will be added to the dataset, representing the cluster assignment of each instance.
6. Switch to the Cluster tab to review EM's clustering results and observe:
    a. EM typically identifies 3 clusters, which correspond closely to the three Iris species.
    b. Review the cluster means and standard deviations for each attribute to understand how the clusters differ.
7. Filter Out Suspected Outliers or Misclassified Instances
    a. Return to the Preprocess tab.
    b. Click Choose under the Filter section and select:
        i. unsupervised → instance → RemoveWithValues
    c. In the RemoveWithValues configuration:
        i. Set Attribute index to the newly created cluster attribute (usually the last attribute).
        ii. Set Nominal value index to the cluster number(s) you wish to remove (for example, if a cluster contains very few instances or shows unusual attribute averages).
    d. Click Apply to remove the selected instances.
8. Compare Dataset Size Before and After Filtering
    a. Check the number of remaining instances in the status bar at the bottom of the Preprocess tab.
    b. Compare this with the original 150 instances in iris.arU' to determine how many were removed.


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open URL...
Open DB...
Generate...
Undo
Edit...
Save...
Filter
Choose
AddCluster -W "weka.clusterers.EM -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1
Apply
Stop
Current relation
Relation: iris-weka.filters.unsupervised.attri...
Attributes: 6
Instances: 150
Sum of weights: 150
Selected attribute
Name: cluster
Type: Nominal
Missing: 0 (0%)
Distinct: 5
Unique: 0 (0%)
No. Label Count Weight
1 cluster1 28 28
2 cluster2 35 35
3 cluster3 42 42
4 cluster4 22 22
5 cluster5 23 23
Attributes
All
None
Invert
Pattern
No. Name
1 sepallength
2 sepalwidth
3 petallength
4 petalwidth
5 class
6 cluster
Class: cluster (Nom)
Visualize All
28
35
42
22
23
Remove
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
EM -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Nom) cluster
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
10:00:45 - EM
Clusterer output
cluster3 1.023 1 1.0175 18.9553 25.0042
cluster4 1 1.0674 1 1 1
cluster5 23.9765 1 1 1 1.0235
[total] 27.9995 32.9688 39.9555 23.0173 29.0277
Time taken to build model (full training data) : 0.28 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 23 (15%)
1 28 (19%)
2 35 (23%)
3 18 (12%)
4 24 (16%)
5 22 (15%)
Log likelihood: -1.81255
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
EM -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Nom) cluster
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
10:00:45 - EM
10:16:37 - EM
Clusterer output
cluster2 26.3151 10.6849 1 1 1
cluster3 1 1 1 1 1
cluster4 1 1 1 1.0674 22.9326
cluster5 1 1 24 1 1
[total] 30.3151 14.6849 28 32.9688 27.0312
Time taken to build model (full training data) : 0.09 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 26 (24%)
1 9 (8%)
2 23 (21%)
3 28 (26%)
4 22 (20%)
Log likelihood: -1.208
Status
OK
Log
x 0


Current relation
Relation: iris-weka.filters.unsupervised.attribute....
Attributes: 6
Instances: 108
Sum of weights: 108
Attributes
All
None
Invert
Pattern
No.
Name
1 sepallength
2 sepalwidth
3 petallength
4 petalwidth
5 class
6 cluster

Reflection:

- What is the purpose of the newly created attribute “Cluster”? (**10pts**)
  - The purpose of creating a cluster and a newly created cluster is to group data to learn patterns and its measurements. This simply means that “Based on the data, this object belongs to this group.” In regards to the iris dataset, we see that objects are flowers so this means we can group these flowers.

    Using EM clustering we can base that “a certain flower belongs to this group.”

- After completing all the procedures, did you encounter the label “Number of clusters selected by cross validation:”? What does this value represent? (**10pts**)
  - This value represents the number of clusters the algorithm decided to use. It shows how the algorithm decides which is the best number of clusters you can use by splitting the data into folds.

    We can even see the log-likelyhood on this part that measures how well the model fits the data. This simply means that the value closer to 0 is ideal.

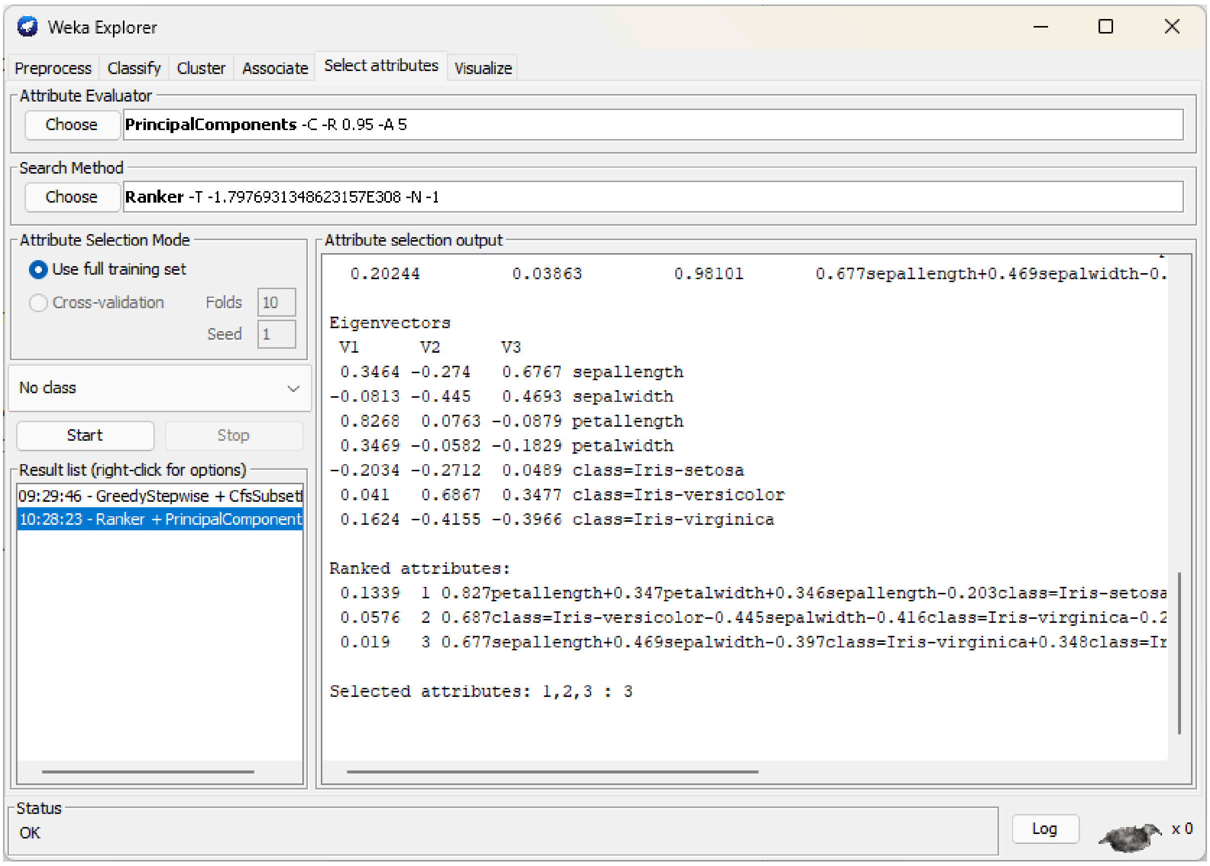
Task 3: Feature Transformation

Instructions:

Perform Principal Component Analysis (PCA) on a sample dataset to reduce dimensionality and transform original attributes into a set of uncorrelated principal components.

1. Launch WEKA and click on Explorer from the main interface.
2. Load the iris.arff dataset
3. Click the Select attributes tab in the Explorer window.
4. Choose PCA as Attribute Evaluator, under Attribute Evaluator, click Choose → select unsupervised → attribute → PrincipalComponents.
5. Configure PCA
   a. Click on PrincipalComponents to open its configuration pane.
   b. Review options:
      i. centerData → keeps data centered before performing PCA (default: True)
      ii. varianceCovered → percentage of variance to retain (default: 0.95 = 95%).
      iii. maximumAttributeNames → controls how many PCs are created (default is automatic).
   c. Leave defaults unchanged for a first run. Click OK.
6. Select a Search Method, under Search Method, click Choose → select Ranker. (This will rank the newly created principal components.)
7. Run the PCA
   a. Click Start to execute the process.
   b. WEKA will generate new principal components and display them with their corresponding variance.
8. Review the Results
   a. Look at the Attribute selection output pane:
      i. You will see the percentage of variance explained by each principal component (V1, V2, etc.).

ii. Verify that most of the variance is captured in the first few PCs, confirming dimensionality reduction.



Reflection:

- How many principal components were created, and what percentage of total variance do they explain? (**15pts**)
  - There are 3 mainly PCs(personal components) were created.
    The percentage of total variance they explain is:

Personal Component	Proportion	Cumulative
PC1	0.86605	0.86605
PC2	0.07633	0.94238
PC3	0.03863	0.98101

  - This shows that 98.10% of the total variance of the three selected components are explained. This means that almost all the information from the original features is preserved.

" I affirm that I have not given or received any unauthorized help in this assignment, and that this work is my own "

Kenneth Arias